

Wild-Type Displacement Model

Mutant HSCs

$x(t)$

grows

fitness $s > 0$

Wild-type HSCs

$N_W(t)$

neutral

rate $\propto F$

fills free space

Free niche

$F(t)$

$E = x + N_W + F$

System of ODEs:

$$\frac{dx}{dt} = s x \left(1 - \frac{x + N_W}{E} \right)$$

$$\frac{dN_W}{dt} = -\frac{dx}{dt} \cdot \frac{B N_W}{F + B N_W}$$

$$\frac{dF}{dt} = -\frac{dx}{dt} \cdot \frac{F}{F + B N_W}$$

$B \rightarrow 0$: expand into free niche only

$B \rightarrow \infty$: displace wild-type directly

Observed VAF:

$$V(t) = \frac{x(t)}{2(N_W(t) + x(t))}$$

VAF trajectories: displacement model vs exponential ($s = 0.3$)

Blues: vary E ($B = 2.5$) · Greens (dashed): vary B ($E = 6 \times 10^5$)

